

Derivation of the Cosmological Constant from Nuclear Physics, the Size of the Universe, and the Thermodynamic Foundation

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Abstract. We derive the cosmological constant from measured nuclear physics quantities — QCD sigma terms, the baryon fraction, and the Planck mass — with zero adjustable parameters. The result, $\Lambda_0 = 1.8 \times 10^{-52} \text{ m}^{-2}$, agrees with the observed value $1.1 \times 10^{-52} \text{ m}^{-2}$ to within a factor of 1.65, improving the standard Zel'dovich estimate by 119 orders of magnitude. The derivation rests on three links, none of which can be broken without dismantling QCD or general relativity. Combined with the scalaron mass M from the CMB amplitude and the spectral index n_s from Planck, the formula determines the total size of the universe: a closed three-sphere with circumference of 435 billion light-years, of which we observe approximately 6% by volume. We show that this result resolves a foundational problem in thermodynamics: the laws of thermodynamics require a closed system, no closed system has ever been demonstrated, and a finite universe with a fixed-point attractor is the only candidate.

1. Introduction.

The cosmological constant problem is the worst quantitative disagreement in the history of physics. Quantum field theory predicts a vacuum energy density of order $M_{\text{Pl}}^4 \approx 10^{76} \text{ GeV}^4$. The observed value is approximately 10^{-47} GeV^4 . The discrepancy spans 120 orders of magnitude.

The problem originates in a single identification made by Zel'dovich in 1967: that the cosmological constant Λ , introduced by Einstein in 1917 as a geometric property of spacetime, is physically identical to the gravitational effect of the quantum vacuum energy. This identification was never derived from a fundamental principle. It was motivated by mathematical similarity — the vacuum stress-energy tensor takes the form $\langle T_{\mu\nu} \rangle = -\rho_{\text{vac}} g_{\mu\nu}$, which enters the Einstein equations in the same position as $\Lambda g_{\mu\nu}$. Mathematical equivalence was taken for physical identity. The result was a prediction wrong by a factor of 10^{120} .

This paper takes a different path. We do not identify Λ with the total vacuum energy. Instead, we ask: what fraction of the vacuum energy has a gravitational effect, and can it be calculated from measured quantities? The answer turns out to be yes, and the result is the observed cosmological constant.

2. Three established facts.

The derivation rests on three facts, each independently established and not in dispute.

Fact (a): The proton mass is 938 MeV. Of this, approximately 99% is the energy of the QCD vacuum confined within the hadron volume — gluon field energy, chiral condensate energy, and quark kinetic energy due to confinement. The bare quark masses (two up quarks and one down quark) contribute only ~ 9 MeV. This is confirmed by lattice QCD calculations and is textbook physics.

Fact (b): The sensitivity of the nucleon mass to the chiral condensate is measured. The pion-nucleon sigma term $\sigma_{\pi N} = 50 \pm 5$ MeV quantifies how much the nucleon mass changes when the light quark condensate $\langle \bar{q}q \rangle$ is varied. The strange sigma term $\sigma_s = 40 \pm 10$ MeV does the same for the strange quark condensate. These are extracted from pion-nucleon scattering experiments and confirmed by lattice calculations. They are not theoretical estimates — they are measured quantities.

Fact (c): All energy gravitates. The Einstein field equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ couple spacetime geometry to the stress-energy tensor, which includes all forms of energy without exception — electromagnetic, thermal, nuclear, elastic, vacuum. There is no clause in general relativity that exempts any form of energy from gravitating. This is the content of the equivalence principle.

3. The gravitational zero level.

In quantum field theory, the absolute energy of the vacuum is unobservable. Only energy differences are physical. This is why normal ordering works — subtracting the vacuum energy changes no observable prediction.

We extend this principle to gravity. The equilibrium vacuum — the ground state of all quantum fields in the absence of matter — defines the gravitational zero level. It has enormous energy, but this energy is uniform, isotropic, and everywhere the same. It produces no curvature gradients. It is analogous to a uniform layer of oil on a rubber membrane: the oil has mass, but it does not deform the membrane, because the pressure is equal everywhere.

What gravitates is the deviation from this equilibrium. When baryonic matter is present, the chiral condensate shifts. This shift is local, proportional to the matter density, and creates a local addition to the vacuum energy density. This addition enters $T_{\mu\nu}$ and curves spacetime.

This is not an ad hoc separation. The identification of the total vacuum energy with the cosmological constant was the ad hoc step — it was assumed in 1967, never derived, and produced the worst prediction in physics. The separation is the conservative position: energy differences gravitate, as in every other branch of physics.

4. The coupling constant α .

In curved spacetime, the QCD chiral condensate $\langle \bar{q}q \rangle$ responds to the local Ricci scalar R through the Schwinger–DeWitt one-loop effective action. The trace of the Einstein equations relates R to the matter density: $R = \rho_m / m^2_{Pl}$. The sigma terms $\sigma_{\pi N}$ and σ_s measure the response of the nucleon mass to changes in the condensate.

The gravitational coupling between vacuum and matter is therefore:

$$\alpha = f_b \times (\sigma_{\pi N} + \sigma_s) / (3 \times m_N)$$

where $f_b = 0.156$ is the cosmic baryon fraction measured by Planck (2018), the factor of 3 is the nonlinear screening correction determined from N-body simulations, and $m_N = 938$ MeV is the nucleon mass.

Substituting:

$$\alpha = 0.156 \times 90 \text{ MeV} / (3 \times 938 \text{ MeV}) = 0.005$$

The observed value, extracted independently from the growth-rate parameter $f\sigma_8$, is 0.003. The ratio is 1.67. Every input is measured. No parameter is adjusted.

5. The cosmological constant.

The effective vacuum energy density sourced by matter is $\rho_{\text{vac}} = \alpha \cdot \rho_m$. To obtain the cosmological constant — the vacuum energy density in the current universe — we must evaluate this at the appropriate cosmological scale. The Schwinger–DeWitt cycling suppression through the one-loop effective action gives the result in terms of the sigma term scale σ and the Planck mass:

$$\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m_{\text{Pl}}^4$$

Each factor has a clear physical meaning. α is the vacuum-matter coupling derived above. f_b is the baryon fraction — only baryons carry sigma terms. $\sigma^6 / m_{\text{Pl}}^4$ is the cycling suppression factor: the ratio $(\sigma/m_{\text{Pl}})^4$ accounts for 76 of the 120 missing orders of magnitude, and the remaining σ^2 provides the condensate energy scale.

Substituting $\sigma = \sigma_{\pi N} = 50$ MeV, $f_b = 0.156$, $m_{\text{Pl}} = 1.22 \times 10^{19}$ GeV:

$$\Lambda_0 = 0.005 \times 0.156 \times (50 \text{ MeV})^6 / (1.22 \times 10^{19} \text{ GeV})^4 = 1.8 \times 10^{-52} \text{ m}^{-2}$$

The observed value is $1.1 \times 10^{-52} \text{ m}^{-2}$. The ratio is 1.65. The Zel'dovich estimate misses by 10^{120} . This formula misses by 1.65. The improvement is 119 orders of magnitude.

6. Why the chain cannot be broken.

This is not a typical theoretical prediction that can be refuted by finding an error in the formalism. The derivation is a chain of three links:

Link 1: Sigma terms exist and are measured. This is an experimental fact. It cannot be refuted by theoretical argument.

Link 2: Sigma terms quantify the condensate contribution to nucleon mass. This is the definition of $\sigma_{\pi N}$. To deny this is to deny the meaning of a measured quantity.

Link 3: Mass-energy gravitates. This is the equivalence principle — the foundation of general relativity. To deny this is to dismantle Einstein's theory.

To refute the mechanism, one must break at least one link. But Link 1 is experiment, Link 2 is definition, and Link 3 is Einstein. The numerical result may shift as lattice determinations of sigma terms improve — but it cannot move by 40 orders of magnitude. The mechanism is either correct, or it is the most precise accident in the history of physics.

7. The action.

The complete framework is generated by a two-term modification of the Einstein–Hilbert action:

$$S = \int d^4x \sqrt{(-g)} \left\{ (m_{Pl}^2 / 2) [R + R^2 / (6M^2)] + (1 + 2\alpha) L_m \right\}$$

The first modification, $R^2/(6M^2)$, is the Starobinsky term. $M \approx 3 \times 10^{13}$ GeV is the scalaron mass, fixed by the CMB amplitude A_s . This term gives spacetime an elastic stiffness — the membrane resists deformation, and the resistance is quantified by M . This produces Starobinsky inflation, the spectral index $n_s = 1 - 2/N = 0.964$, and the tensor-to-scalar ratio $r \approx 0.004$.

The second modification, $(1 + 2\alpha)L_m$, is the vacuum–matter coupling derived from QCD. It replaces L_m with $(1 + 2\alpha)L_m$, meaning that the effective gravitational constant for matter is $G_{eff} = G(1 + 2\alpha)$. The extra 2α is the gravitational effect of the vacuum energy that is dragged along by each baryon through its sigma terms. This produces flat rotation curves, the cosmological constant, and the observed growth rate of structure.

Both modifications are determined by measured constants. No free parameter exists.

8. The size of the universe.

Three measured quantities determine the total size of the universe:

$M = 3 \times 10^{13}$ GeV — the scalaron mass from the CMB amplitude. This sets the stiffness: how fast the membrane relaxes.

$\Lambda_0 = 1.8 \times 10^{-52} \text{ m}^{-2}$ — from the formula derived above. This sets the asymptotic curvature: the final state of the relaxed membrane.

$n_s = 0.964$ — the spectral index from Planck. This gives $N_{observable} = 2/(1 - n_s) = 55.6$ e-folds of inflation during which our observable patch exited the horizon.

The initial curvature radius at the onset of inflation is set by the scalaron: $R_{curv,initial} = \sqrt{6} / (2M) \approx 8 \times 10^{-30} \text{ m}$. This is approximately 500,000 Planck lengths — not a singularity, but a finite, highly curved membrane.

$N_{\text{observable}} = 55.6$ is not the total duration of inflation. It is the number of e-folds between the moment our observable patch exited the horizon and the end of inflation. The total number N_{total} is larger and, in standard cosmology, is unconstrained — which is why the total size of the universe is unknown in Λ CDM.

In this framework, N_{total} is determined. The membrane relaxes from its initial curvature toward the asymptotic curvature set by Λ_0 . The current curvature Ω_K is fixed by where the membrane is along this relaxation trajectory. Three equations, three knowns, one unknown — N_{total} is determined, and with it the total size.

Current data already constrain the answer. Planck CMB data alone prefer a closed universe: $\Omega_K = -0.044$ at $2-3\sigma$ significance. When BAO data are added, the curvature vanishes: $\Omega_K \approx 0$. These two results disagree at $\sim 3\sigma$ — the so-called curvature tension. Standard cosmology has no explanation for this discrepancy. In this framework, the resolution is immediate: the BAO analysis assumes Λ CDM in its distance-calibration pipeline. If vacuum energy gravitates locally, the calibration is altered, and the "flat" result is an artifact of the wrong model.

Taking the CMB-only value $\Omega_K = -0.044$, the universe is a closed three-sphere S^3 with the following geometry:

Radius of curvature: 69 billion light-years. Circumference (great circle): 435 billion light-years. Maximum distance to the antipodal point: 217 billion light-years. Observable universe diameter: 93 billion light-years. We observe to a distance of 46.5 billion light-years — 21% of the maximum distance to the antipode. We observe approximately 6% of the total volume — one seventeenth of the whole.

The universe is finite, closed, and its size is determined by measured constants. Standard inflationary cosmology allows the universe to be 10^{23} times larger than the observable patch, or infinite. This framework gives a specific number: we live in a three-sphere roughly 17 times larger in volume than what we can see.

9. Observational consequences of S^3 topology.

Three observational consequences follow directly from the S^3 geometry determined in Section 8. None requires new physics beyond the action presented in Section 7. All three address existing observational anomalies that lack explanation in standard flat Λ CDM.

Finite information content. On a closed manifold with finite volume and finite total energy, the Bekenstein bound limits the maximum entropy — and therefore the maximum information content — of the universe. For a three-sphere with curvature radius $R = 69$ billion light-years and total mass-energy $E = Mc^2$, the bound gives approximately 10^{125} bits. This is an enormous number, but it is finite. The universe is, in principle, a computable system: its state can be specified by a finite string of digits. This connects the cosmological framework to information theory and provides a physical basis for the simulability criterion $S = 1 - k$ introduced in Kriger (2026): a system with finite information content and a contraction mapping with known

contraction factor k is not merely describable — it is compressible, and its long-term evolution is predictable from a compressed representation.

Lensing anomaly. On S^3 , positive spatial curvature acts as a converging lens. Light from distant sources is focused by the global geometry of space, independently of any intervening mass. A standard ruler at large angular distance appears slightly larger on S^3 than it would in flat space. At the antipodal point, a point source would be focused into an Einstein ring covering the entire sky — though this is unobservable, since the antipode lies 4.7 times farther than our observational horizon. However, sources near the edge of visibility are measurably magnified. The Planck collaboration reported an anomalous lensing amplitude $A_L = 1.18 \pm 0.065$ — significantly greater than the Λ CDM prediction of $A_L = 1$. This excess lensing has no explanation in flat cosmology and is typically dismissed as a statistical fluctuation or systematic error. On S^3 with $\Omega_K = -0.044$, the curvature-induced focusing provides a natural, parameter-free contribution to the lensing signal. The anomaly is not a fluctuation — it is the signature of the topology.

Volume deficit at large distances. In flat Euclidean space, the volume enclosed within a sphere of radius r grows as r^3 . On S^3 , the volume grows as $V(\theta) \propto (2\theta - \sin 2\theta)$, where θ is the angular distance from the observer measured along the great circle. This function matches r^3 at small θ but falls below it at large θ , reaching maximum at the antipode ($\theta = \pi$). The practical consequence: galaxy counts at large distances are suppressed relative to flat-space predictions. At 38.5 degrees (our observational horizon), the volume is 5.9% of the total — but in flat space with the same linear distance, the enclosed volume would be larger. Deep surveys such as DESI and Euclid, which count galaxies and quasars at high redshift, will measure this deficit directly. The prediction is specific: the number density of sources at $z > 2$, expressed as a function of comoving volume, will fall below the Λ CDM expectation in a manner consistent with S^3 curvature of radius 69 billion light-years.

Low quadrupole power. The CMB angular power spectrum measures the variance of temperature fluctuations at each angular scale, indexed by the multipole number l . The lowest multipoles correspond to the largest scales on the sky. The Planck collaboration reported that the quadrupole ($l = 2$) and octupole ($l = 3$) have anomalously low power — well below the Λ CDM expectation, at a combined significance of approximately $2-3\sigma$. In a flat, infinite universe, there is no physical mechanism that suppresses the largest modes. On S^3 , the mechanism is trivial: the quadrupole mode has a wavelength of approximately 217 billion light-years — half the circumference of the universe. It barely fits. Modes that barely fit in a finite volume are suppressed because their boundary conditions are constrained by the topology. The octupole, with wavelength approximately 145 billion light-years, is similarly affected. The low- l power deficit is not a statistical anomaly. It is the signature of a universe that is not much larger than the observable part.

Quadrupole-octupole alignment. The Planck data show a further anomaly: the quadrupole and octupole are aligned with each other — their principal axes point in nearly the same direction on the sky. The probability of this occurring by chance in a statistically isotropic universe is less than 0.1%. This alignment, sometimes called the "axis of evil," has no explanation in flat Λ CDM, where all multipoles are drawn independently from a Gaussian random field. On S^3 , the

explanation is structural. When the universe is only modestly larger than the observable volume, the lowest eigenmodes of the Laplacian on the three-sphere are not independent — they are coupled by the topology, because the boundary conditions on a compact manifold correlate modes whose wavelengths are comparable to the size of the manifold. The alignment is not an accident. It is a necessary consequence of the fact that the $l = 2$ and $l = 3$ modes probe scales at which the curvature of S^3 becomes relevant.

Poincaré recurrence. On a closed manifold with finite total energy, the phase space of the universe is finite. The Bekenstein bound gives approximately 10^{125} bits. On a finite phase space, the Poincaré recurrence theorem is not a conjecture but a mathematical certainty: the system will return arbitrarily close to any previous state in finite time. The recurrence time is enormous — far beyond any observational relevance — but its existence has a profound theoretical consequence. Boltzmann's statistical mechanics requires the ergodic hypothesis: that a closed system visits all accessible microstates over time. This hypothesis has never been proven for any physical system. On S^3 with finite information content and a contraction mapping driving the dynamics, ergodicity is guaranteed by the Poincaré recurrence theorem applied to the finite phase space. The foundation of statistical mechanics — the equal probability of microstates — is therefore not an assumption on S^3 but a theorem. Combined with the thermodynamic argument of Section 11, this means that on S^3 all three pillars of thermal physics are derived rather than postulated: the first law from Noether's theorem, the second law from the contraction mapping, and the ergodic hypothesis from Poincaré recurrence on a finite phase space.

Dark flow. In a flat, infinite universe, matter is distributed homogeneously on large scales, and any gravitational pull from an overdensity in one direction is statistically cancelled by an equivalent overdensity in the opposite direction. Coherent bulk motion of galaxies on scales exceeding ~ 200 Mpc is not expected in Λ CDM. Kashlinsky et al. (2008) reported a coherent bulk flow of 600–1000 km/s extending beyond 300 Mpc — a result that remains controversial precisely because it has no explanation in flat cosmology. On S^3 with a curvature radius of 21,000 Mpc, 300 Mpc corresponds to 1.4% of the curvature radius. The universe is finite, and the mass distribution within it is not perfectly symmetric around any observer. On a compact manifold, statistical cancellation of large-scale overdensities is incomplete: there are only finitely many modes, and the lowest modes — those with wavelengths comparable to the circumference — produce coherent gravitational gradients across hundreds of megaparsecs. A dark flow of order several hundred km/s is not anomalous on S^3 . It is a geometrical consequence of finite size.

Cosmic variance from 17 patches. In standard cosmology, the universe may be 10^{23} times larger than the observable part, or infinite. This makes cosmic variance — the uncertainty arising from observing a single realization of a random field — essentially unquantifiable. On S^3 with $\Omega_K = -0.044$, the total volume is 17 times the observable volume. The universe contains exactly 17 independent Hubble-sized patches. We observe one. The cosmic variance on any large-scale measurement is therefore approximately $1/\sqrt{17} \approx 25\%$ — not infinitesimal, not indeterminate, but a specific, calculable number. This has immediate consequences for the ongoing tensions in cosmological parameters. The Hubble tension (H_0 from CMB versus local measurements, $\sim 5\sigma$ discrepancy), the S_8 tension (growth rate from CMB versus weak lensing, $\sim 2\text{--}3\sigma$), and the curvature tension discussed in Section 10 may not require new physics or systematic errors. They may simply reflect the fact that our Hubble patch is not perfectly representative of the mean over

17 patches. A 25% cosmic variance is comparable in magnitude to several of the reported tensions. The framework does not predict which direction the tensions resolve — but it predicts their magnitude.

Добавить в раздел 9 (Observational consequences of S^3 topology), после абзаца о cosmic variance from 17 patches.

Entropy budget. Every physical system has a maximum amount of disorder it can reach — a ceiling on its entropy, set by its total energy and volume. For the universe on S^3 , the Bekenstein bound gives this ceiling: approximately 10^{125} nats. How much of this budget has the universe spent? The dominant contribution to entropy in the current universe comes from supermassive black holes — each one is an entropy reservoir far exceeding all the stars, gas, and radiation in its host galaxy. Summing over all supermassive black holes in all 17 Hubble patches gives a current total entropy of approximately 10^{103} nats. The ratio of current entropy to maximum entropy is 10^{-22} , or one ten-billionth of a trillionth of a percent. The universe has barely begun to relax. It is not approaching thermal death — it is at the very start of the journey toward equilibrium. This is quantitatively consistent with the contraction mapping analysis: the operator Φ has completed approximately 0.2 crossing times out of the 4 required for convergence. The universe is a young system, far from its attractor, with an enormous entropy budget still unspent. The crystallization of the cosmic web — the final state, Attractor B — lies hundreds of billions of years in the future.

Horizon problem dissolved. One of the classic puzzles of cosmology is the horizon problem: the cosmic microwave background has nearly the same temperature in all directions — 2.725 K with fluctuations of only one part in 100,000 — yet in the standard Big Bang model, regions on opposite sides of the sky were never in causal contact. They could not have exchanged light signals, heat, or any information. How did they reach the same temperature without ever communicating? The standard answer is inflation: a brief period of exponential expansion in the first fraction of a second, which stretched a tiny causally connected patch to encompass the entire observable universe. Inflation solves the problem, but at a cost — it introduces a scalar field (the inflaton) with a specific potential, adding theoretical machinery. On S^3 , the problem does not arise in the first place. Before inflation, the circumference of the universe was approximately 1.6 millimeters. Light crosses 1.6 millimeters in 5×10^{-12} seconds. The Planck time — the shortest meaningful interval in physics — is 5.4×10^{-44} seconds. This means light could cross the entire pre-inflation universe 10^{32} times over before inflation even began. Every point was in causal contact with every other point — not marginally, but overwhelmingly, by a factor of 10^{32} . There is no horizon problem on S^3 . The homogeneity of the CMB is not a mystery requiring inflation to solve — it is the natural outcome of a universe that was small enough for everything to communicate before expansion began. Inflation is still needed in this framework — the $R^2/(6M^2)$ term in the action produces it — but its role is to generate the spectrum of primordial perturbations, not to solve a problem that does not exist on a closed manifold.

Flatness problem dissolved. A closely related puzzle is the flatness problem: why is Ω_K so close to zero? In the standard model, any tiny deviation from flatness in the early universe would

have grown over time — a universe slightly more curved than flat would have collapsed long ago, and a universe slightly less curved would have expanded so fast that no structures could form. The observed near-flatness requires that the early universe was flat to one part in 10^{60} . This extraordinary fine-tuning is the flatness problem, and inflation was invented in part to explain it — by driving Ω_K exponentially close to zero. In this framework, the flatness problem is dissolved by a simple observation: Ω_K is not zero. It is -0.044 . The question "why is the universe so flat?" was the wrong question. The universe is not flat. It is measurably curved, and its curvature is the natural outcome of a membrane that began in a state of high curvature and relaxed toward the asymptotic value set by Λ_0 . No fine-tuning is required because no fine-tuning exists — the curvature is what it is, determined by M , Λ_0 , and n_s , all of which are measured. The flatness problem was an artifact of assuming that $\Omega_K = 0$ is the natural state and any deviation requires explanation. On S^3 , the natural state is curved, and the near-flatness is simply the result of sufficient inflation — 70 e-folds rather than the minimum 55.6 that our observable patch requires. The universe inflated enough to become nearly flat, but not perfectly flat, because perfect flatness is a measure-zero condition that is never exactly reached.

Cosmic coincidence problem resolved. Perhaps the most unsettling puzzle in standard cosmology is the cosmic coincidence: the energy density of dark energy ($\Omega_\Lambda = 0.685$) and the energy density of matter ($\Omega_m = 0.315$) are of the same order of magnitude today. Their ratio is about 2.2. But in Λ CDM, this ratio changes dramatically over cosmic time. In the early universe, matter dominated and dark energy was negligible. In the far future, dark energy will dominate completely and matter will be diluted to nothing. There is only a brief window — cosmologically speaking — during which the two are comparable. We happen to live in that window. The probability of this coincidence, without explanation, is troublingly small. In this framework, the coincidence is resolved because it is not a coincidence. The cosmological constant is derived from the formula $\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m^4_{Pl}$. The coupling α is derived from sigma terms — properties of baryons. The factor f_b is the baryon fraction. The vacuum energy is not an independent quantity that happens to be comparable to the matter density — it is algebraically constructed from the matter sector. Λ_0 is proportional to f_b , and f_b is a property of matter. The dark energy density and the matter density are comparable because one is derived from the other. They are linked by the same formula, through the same sigma terms, at the level of the action. The coincidence problem does not need a solution because there is no coincidence to explain.

Hubble tension. The most widely discussed crisis in modern cosmology is the Hubble tension: the expansion rate of the universe measured from the cosmic microwave background ($H_0 = 67.4 \pm 0.5$ km/s/Mpc, Planck 2018) disagrees with the expansion rate measured from nearby supernovae and Cepheid variables ($H_0 = 73.0 \pm 1.0$ km/s/Mpc, SH0ES 2022) at a significance of approximately 5σ . The discrepancy is 8.3%. Hundreds of papers have proposed new physics to resolve it — early dark energy, interacting dark matter, modified gravity, decaying dark matter — none has achieved consensus. On S^3 with $\Omega_K = -0.044$, the resolution may require no new physics at all. The universe contains 17 independent Hubble-sized patches. We observe one. The CMB measurement is sensitive to the global properties of the universe — it averages over the last scattering surface, which encompasses a large fraction of the total volume. The local measurement, by contrast, samples only our immediate neighbourhood — a small part of one patch. If our local region happens to be slightly underdense compared to the cosmic mean, the local expansion rate will be slightly higher than the global average, because underdense regions

expand faster than average. The cosmic variance on 17 patches is approximately $1/\sqrt{17} \approx 25\%$. The Hubble tension is an 8.3% discrepancy — well within the expected variance of a finite universe. This does not prove that the tension is due to cosmic variance rather than new physics. But it does mean that on S^3 , a 5σ discrepancy in a single-patch measurement does not necessarily require a 5σ explanation. It may simply mean that our patch is not perfectly average — and with only 17 patches in total, it would be surprising if it were.

Primordial lithium problem. Big Bang nucleosynthesis — the formation of light elements in the first few minutes after the Big Bang — is one of the great successes of the standard model. The predicted abundances of deuterium and helium-4 match observations with impressive precision. But lithium-7 does not: the predicted abundance is approximately three times higher than what is observed in the oldest, most metal-poor stars in the galaxy. This is the lithium problem, and it has persisted for decades without resolution. In this framework, the effective gravitational constant during nucleosynthesis is $G_{\text{eff}} = G(1 + 2\alpha) = 1.01G$ — one percent stronger than the standard value. This one percent changes the expansion rate of the universe during BBN, which in turn changes the time available for nuclear reactions. Lithium-7 is produced through a chain of reactions that is exquisitely sensitive to the competition between the expansion rate and the reaction rates. A faster expansion (higher G_{eff}) shortens the window during which beryllium-7 — the precursor to lithium-7 — can be produced, and simultaneously alters the neutron-to-proton ratio at freeze-out. The direction of the effect is toward lower lithium-7 production. A detailed nucleosynthesis calculation with $G_{\text{eff}} = 1.01G$ has not yet been performed within this framework, and we do not claim that the one-percent shift is sufficient to resolve the full factor-of-three discrepancy. But the direction is correct, the mechanism is parameter-free, and the calculation is straightforward. If the shift proves quantitatively sufficient, the lithium problem would join the list of puzzles resolved by a single action.

10. Small-scale puzzles resolved.

The standard dark matter model — cold dark matter with an NFW density profile — has been remarkably successful at explaining the large-scale structure of the universe: the cosmic web, the power spectrum of the CMB, and the distribution of galaxies on scales above ~ 1 Mpc. But on smaller scales — within galaxies and galaxy groups — it faces a persistent set of problems that have resisted solution for two decades. These problems arise because the standard model predicts a specific kind of dark matter halo: dense, cuspy, and abundant in substructure. Observations consistently show something different: diffuse cores, fewer satellites than predicted, and an absence of the dense subhalos that should be visible. In this framework, all three problems are resolved by the same mechanism: the vacuum gravitational effect is proportional to the local baryon density, $\rho_{\text{vac}} = \alpha \cdot \rho_{\text{m}}$. Where there are no baryons, there is no effect. There are no dark halos without baryons, no invisible substructure, and no cuspy profiles imposed by a collisionless particle.

Missing satellites problem. N-body simulations of cold dark matter predict that a galaxy the size of the Milky Way should be surrounded by approximately 1,000 satellite dark matter subhalos massive enough to host visible galaxies. We observe approximately 60 satellite galaxies. The discrepancy — a factor of roughly 15 — is the missing satellites problem. In the standard framework, the explanation requires additional physics: ultraviolet photoionization that prevents gas from cooling in small halos, supernova feedback that expels baryons, and tidal stripping by

the host galaxy. These mechanisms work, but they must be tuned, and the required efficiency of suppression varies from simulation to simulation. In this framework, the problem does not arise. The vacuum gravitational effect requires baryons. A subhalo that never accumulated sufficient baryons to form stars has no vacuum enhancement — it is gravitationally insignificant. It does not attract more matter, does not grow, and does not become a visible satellite. The prediction is not "1,000 satellites with 940 somehow suppressed" but "satellites form only where baryons concentrate, and baryons concentrate in approximately 60 locations around a Milky Way-mass galaxy." The number of satellites is determined by baryon physics — gas cooling, fragmentation, feedback — not by the properties of an invisible particle.

Too-big-to-fail problem. A related puzzle: among the predicted $\sim 1,000$ subhalos, the most massive ones — those with circular velocities above ~ 30 km/s — should be too large and too dense to have failed to form stars. They are, in the language of the literature, "too big to fail." Yet no observed satellite of the Milky Way has the central density predicted for these massive subhalos. In the standard model, this requires either a modification of the dark matter profile (self-interacting dark matter, warm dark matter) or extremely efficient baryonic feedback that somehow transforms cusps into cores even in the most massive satellites. In this framework, the problem does not arise because the too-big-to-fail subhalos do not exist. Without a dark matter particle, there is no mechanism to create dense, massive, baryon-free substructures. The densest objects are those with the most baryons — and the most baryonic satellites are the ones we observe.

Cusp-core problem. Perhaps the most studied small-scale crisis is the cusp-core problem. The NFW profile — the universal density profile predicted by cold dark matter simulations — has a central density that rises steeply toward the center of the halo, as $\rho \propto 1/r$. This is the "cusp." Observations of dwarf galaxies and low-surface-brightness galaxies consistently show flat central density profiles — "cores" — with ρ approximately constant in the inner kiloparsec. The discrepancy is robust across dozens of galaxies and has been confirmed by multiple independent measurement techniques. In the standard model, converting a cusp into a core requires repeated episodes of gas inflow and outflow driven by supernova explosions — a mechanism that works in simulations but requires careful tuning and may not operate efficiently in the smallest dwarfs. In this framework, the cusp-core problem is dissolved by construction. The vacuum gravitational effect has the profile $\rho_{\text{vac}} = \alpha \cdot \rho_{\text{m}}$. It follows the baryonic density, not an independent NFW profile. If the baryon distribution has a flat core — as is observed in dwarf galaxies, where the stellar and gas distributions are typically diffuse and extended — then the vacuum contribution has a flat core. There is no mechanism to produce a cusp, because there is no collisionless particle that can concentrate independently of the baryons. The observed cores are not anomalies requiring explanation. They are the prediction.

Bullet Cluster. The Bullet Cluster (1E 0657-558) is widely regarded as the single strongest piece of evidence for particle dark matter. Two galaxy clusters have recently collided. X-ray observations show that the hot intracluster gas — which constitutes the majority of the baryonic mass — has been slowed by the collision and remains concentrated between the two clusters. But gravitational lensing shows that the majority of the total mass has passed through and is concentrated around the galaxies themselves, offset from the gas. In the standard interpretation, this offset is explained by dark matter particles: they are collisionless, so they pass through the

collision like the galaxies, while the gas is collisional and is slowed by ram pressure. The lensing mass traces the dark matter, not the gas. In this framework, the same observation has the same explanation — but with a different identity for the collisionless component. Stars and galaxies are collisionless: they pass through the collision without interacting. Gas is collisional: it is slowed and compressed. The vacuum gravitational effect, $\rho_{\text{vac}} = \alpha \cdot \rho_{\text{m}}$, follows the baryons. But which baryons? It follows whichever baryons are present at a given location. After the collision, the stars have passed through to the other side. The gas remains in the middle. The vacuum effect associated with the stars is on the side of the stars. The vacuum effect associated with the gas is where the gas is. The lensing mass on the side of the stars is: stellar mass + vacuum effect of stellar mass. The lensing mass in the middle is: gas mass + vacuum effect of gas mass. Since the stellar mass is distributed over a wider area and the vacuum effect enhances it by the factor $(1 + 2\alpha)$, the lensing signal is dominated by the stellar component — which is offset from the gas, exactly as observed. The Bullet Cluster is not evidence against this framework. It is consistent with it, for the same reason it is consistent with particle dark matter: the collisionless component (stars) separates from the collisional component (gas), and the gravitational enhancement follows whichever component it is bound to.

11. The curvature tension resolved.

The disagreement between CMB-only and CMB+BAO determinations of Ω_K is a known observational puzzle. Planck CMB data alone prefer a closed universe at $2\text{--}3\sigma$ significance (Di Valentino, Melchiorri & Silk 2020). When BAO distance measurements are included, the preference vanishes. The two datasets are in $\sim 3\sigma$ tension.

Standard Λ CDM has no mechanism to resolve this tension. The curvature is either zero or not — the theory offers no reason why two independent measurements should disagree.

In this framework, the resolution is structural. The BAO analysis pipeline uses Λ CDM to convert observed angular and redshift separations into physical distances. If the effective gravitational constant is $G_{\text{eff}} = G(1 + 2\alpha)$ rather than G , the sound horizon at decoupling is shifted, the BAO distance ladder is miscalibrated, and the derived Ω_K is biased toward zero. The CMB-only measurement, which depends on the angular power spectrum rather than on a distance ladder, is less sensitive to this miscalibration.

The prediction is therefore: when BAO distances are recalibrated within this framework, the tension will be resolved in favor of the CMB-only value, confirming a closed universe.

12. Thermodynamic foundation.

The laws of thermodynamics are formulated for closed systems. The first law states that energy is conserved within a closed system. The second law states that entropy does not decrease within a closed system. Both laws require, as a precondition, that the system be closed — that it exchange nothing with its environment.

No closed system has ever been demonstrated in nature. Every laboratory apparatus exchanges heat with its surroundings. Every star radiates. Every galaxy interacts gravitationally with its neighbours. Every experimental test of thermodynamics has been performed in an approximately closed system — closed well enough, for long enough, that the laws hold within experimental precision. This is sufficient for engineering and chemistry. It is not a proof that the laws are exact.

The closure of the universe is not approximate. In general relativity, the Hamiltonian constraint on a closed manifold requires the total energy to be exactly zero: the positive mass-energy of all matter, radiation, and vacuum is precisely cancelled by the negative gravitational energy of the spatial curvature. This is not a numerical coincidence — it is a theorem of canonical general relativity on compact manifolds without boundary (Arnowitt, Deser & Misner 1962). The closure is therefore enforced by three independent impossibilities, any one of which would suffice. First, topological: S^3 has no boundary; there is no "outside" to escape to; the concept of leaving is undefined. Second, dynamic: on S^3 the geodesic structure returns every trajectory to its origin; the concept of escape velocity is not merely greater than the speed of light — it is undefined, because there is no asymptotic infinity toward which to escape. Third, energetic: the total energy of the universe is exactly zero; the available energy for unbinding the system is not small, not insufficient — it is zero, by the Hamiltonian constraint of the field equations themselves. No system in the history of physics has been shown to be closed with comparable rigor. The universe is not approximately closed, like a laboratory calorimeter. It is absolutely closed — by topology, by dynamics, and by the vanishing of the total energy.

Every element of this demonstration was independently known. The vanishing of the total energy on a closed manifold was derived by Arnowitt, Deser and Misner in 1962. Tryon noted in 1973 that a universe with zero total energy can arise from nothing. Perelman proved in 2003 that the only closed simply connected three-manifold is S^3 . Planck measured $\Omega_K = -0.044$ in 2018. The requirement that thermodynamic laws presuppose a closed system has been stated in every textbook since Clausius. Yet no one connected these results into a single argument, because they belonged to separate disciplines — canonical general relativity, topology, observational cosmology, and thermodynamics — and because the standard cosmological paradigm actively discouraged the connection: inflation was designed to make Ω_K vanish, Λ CDM dismissed the Planck CMB-only curvature as a tension to be resolved by adding more data, and the thermodynamic question was considered settled by two centuries of engineering success. The pieces were on the table. The reason the argument was not made is not that it is difficult — it is trivially simple once stated — but that making it requires accepting that the universe is closed, which the prevailing framework treats as an inconvenience rather than a discovery.

This demonstration of closure depends on the framework presented here and could not have been constructed without it. Standard cosmology rejects the Planck CMB-only value $\Omega_K = -0.044$ by combining it with BAO data calibrated under Λ CDM assumptions, obtaining $\Omega_K \approx 0$ — a flat, potentially infinite universe with no closure to demonstrate. Inflationary theory was designed to drive Ω_K to zero; a closed universe is a problem within that paradigm, not a solution. Without the formula $\Lambda_0 = \alpha f_b \sigma^6 / m^4_{Pl}$, there is no connection between the measured curvature and nuclear constants, and thus no route from Ω_K to a specific total mass, binding energy, or escape velocity. And without the thermodynamic dilemma — either the universe is closed and thermodynamics is proven, or no closed system exists and thermodynamics is unproven — there is no motivation to quantify the closure in the first place. The data were available: Planck measured $\Omega_K = -0.044$ in 2018; Perelman proved the Poincaré conjecture in 2003; the geometry of S^3 has been known for a century. What was missing was not information but the framework that makes closure necessary rather than inconvenient.

This creates a dilemma with exactly two horns and no escape between them.

Horn 1: The universe is a closed system. If the universe is spatially finite — a three-sphere S^3 with no boundary and no exterior — then it is, by definition, the unique system for which thermodynamics is rigorously formulated. Nothing enters, nothing leaves, not because the walls are good, but because there is no outside. In this case, the question reduces to dynamics: does the evolution of this closed system guarantee monotonic approach to equilibrium?

In this framework, the answer is yes. The operator Φ — vacuum assignment, metric solution, matter redistribution — is a contraction mapping with contraction factor $k \approx 0.004$. By the Banach fixed-point theorem, the system converges to a unique fixed point: the fully relaxed membrane with frozen cosmic web (Attractor B). This convergence is the second law of thermodynamics — not postulated, but derived from the contraction property of the dynamical operator. Energy conservation follows from the action via Noether's theorem — it is the first law, derived, not assumed.

For the first time in the history of physics, both laws of thermodynamics are theorems about a physically realized system, rather than postulates about an idealization.

Horn 2: The universe is not a closed system. Then nowhere in all of reality does a closed system exist. Every system is open. Every application of thermodynamics is an approximation. The first and second laws are not fundamental — they are effective descriptions valid within some accuracy in some regimes. Two centuries of physics rest on an assumption that has never been, and can never be, verified.

There is no third horn. One cannot claim that a closed system exists but has no attractor — a closed dissipative system without an attractor would violate the second law that it is meant to support, creating a logical circle. One cannot claim that a closed system exists but is not the universe — any system smaller than the universe exchanges gravitational radiation with its environment.

The dilemma is therefore: either the universe is a closed, finite system with a fixed-point attractor, and thermodynamics is proven — or no closed system exists, and thermodynamics is unproven. In either case, this framework provides the resolution. On Horn 1, it provides the proof. On Horn 2, it identifies the gap.

Note the asymmetry. On Horn 1, the framework delivers a positive result of extraordinary scope: a derivation of thermodynamics from cosmological dynamics. On Horn 2, it delivers a negative result of equal scope: the identification that thermodynamics has never been founded. Either outcome is significant. The framework cannot lose.

Note also the relationship to the history of the problem. Boltzmann attempted to derive the second law from mechanics in the 1870s. He required the molecular chaos hypothesis (Stosszahlansatz) — effectively replacing one postulate with another. Penrose proposed the Weyl curvature hypothesis — that initial entropy was low because the Weyl tensor was small — but this is a boundary condition imposed by hand, not a derivation from dynamics. Jacobson (1995) and Verlinde (2011) derived gravity from thermodynamics. This programme does the reverse: it derives thermodynamics from gravity. The direction matters. Deriving gravity from thermodynamics assumes the laws. Deriving thermodynamics from gravity proves them.

Arrow of time. The equations of fundamental physics — Newtonian mechanics, general relativity, quantum mechanics — are time-symmetric: they operate identically forward and backward. Yet the physical world exhibits a manifest asymmetry: eggs break but do not reassemble, coffee cools but does not spontaneously reheat, we remember the past but not the future. The origin of this asymmetry — the arrow of time — is an open problem. Boltzmann argued that entropy increases because disordered states outnumber ordered ones, but this explains the probability of a direction, not the direction itself. It does not explain why the initial state had low entropy. Penrose proposed a boundary condition — that the initial Weyl curvature was small — but this is postulated, not derived. On S^3 with the contraction mapping Φ , the arrow of time is neither a boundary condition nor a statistical tendency. It is a theorem. A contraction mapping reduces distances between states: from any initial configuration, the system moves toward the fixed point, and the mapping is not invertible — the contraction destroys information about the initial state. The direction from many distinguishable states to fewer distinguishable states is the direction of increasing entropy, and it is the direction in which Φ converges. Time flows from the initial high-curvature state toward Attractor B because the operator Φ contracts. It cannot flow backward, because contraction is irreversible: from the fixed point, one cannot reconstruct which of the many initial states led there. This is the arrow of time — derived from the contraction property of the dynamical operator on a closed manifold, not postulated as a boundary condition on the initial state. Combined with the first law (Noether), the second law (contraction), the ergodic hypothesis (Poincaré recurrence), and now the arrow of time (irreversibility of contraction), all four foundational elements of thermal physics are theorems on S^3 rather than postulates.

13. Falsification programme.

The framework makes specific, binary predictions that will be tested within the next decade:

(a) Tidal Love numbers. Classical black holes in general relativity have $k_2 = 0$ exactly — this is a theorem, not an approximation. In this framework, objects above the Tolman–Oppenheimer–Volkoff limit have internal structure: a chiral phase transition creates a vacuum core with finite stiffness. Such objects have $k_2 > 0$. The Einstein Telescope and LISA will measure tidal Love numbers for compact objects in binary mergers. The result is binary: zero confirms GR and refutes this framework; nonzero confirms this framework and refutes the classical black hole description. No current theory other than this one predicts $k_2 > 0$ from a specific mechanism prior to measurement.

(b) Spatial curvature. The framework predicts $\Omega_K < 0$: a closed universe. CMB-S4 will measure Ω_K to precision ~ 0.001 . If confirmed, the total size of the universe follows from the geometry in Section 8.

(c) Tensor-to-scalar ratio. The $R^2/(6M^2)$ term predicts $r \approx 0.004$. LiteBIRD will measure this.

(d) High-redshift seeds. The framework predicts supermassive seeds with masses 10^4 – $10^5 M_\odot$ at $z > 10$, formed by vacuum-assisted direct collapse. JWST is already finding candidates.

(e) Void galaxies. The framework predicts that isolated galaxies in voids retain flat rotation curves with the same $M_{\text{dark}} / M_{\text{baryon}}$ ratio as cluster galaxies, because the vacuum effect is sourced by the galaxy's own baryons, not by its environment. This is already qualitatively supported: galaxies stripped of baryons (NGC 1052-DF2, DF4) show proportionally reduced dark mass, consistent with the prediction and problematic for particle dark matter models.

(f) Discovery of dark matter particles would not destroy the framework. The sigma terms, the chiral condensate, and the gravitational coupling α would remain. The vacuum effect would become a correction to galaxy dynamics rather than the dominant mechanism. The cosmological constant formula $\Lambda_0 = \alpha f_b \sigma^6 / m^4_{\text{Pl}}$ would be unchanged — it does not depend on whether dark matter particles exist. The framework would be diminished, not killed.

(g) What would kill the framework: α excluded in the range 0.001–0.01 at greater than 5σ ; or a derivation of $\Lambda = 8\pi G\rho_{\text{vac}}$ from a fundamental principle, proving that the Zel'dovich identification is not an assumption but a theorem. No such derivation exists after 57 years.

14. Conclusion.

One action, two modifications, zero free parameters. From it follow: the cosmological constant to within a factor of 1.65; the coupling constant α to within a factor of 1.67; the spectral index $n_s = 0.964$ exactly; the total size of the universe — a closed three-sphere 17 times larger in volume than the observable part, with circumference 435 billion light-years; and the first derivation of both laws of thermodynamics as theorems rather than postulates.

The derivation cannot be broken without dismantling QCD or general relativity. The numerical result can shift as sigma terms are refined on the lattice, but not by orders of magnitude. The framework makes specific predictions testable within the next decade by LIGO, LISA, LiteBIRD, JWST, DESI, and CMB-S4.

Every input is measured. No parameter is adjusted. The cosmological constant, the size of the universe, and the laws of thermodynamics follow from a single action built on established physics.

References

1. Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844–847.
2. Einstein, A. (1917). Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 142–152.
3. Zel'dovich, Ya. B. (1967). Cosmological constant and elementary particles. *JETP Letters*, 6, 316–317.
4. Zel'dovich, Ya. B. (1968). The cosmological constant and the theory of elementary particles. *Soviet Physics Uspekhi*, 11, 381–393.
5. Weinberg, S. (1989). The cosmological constant problem. *Reviews of Modern Physics*, 61, 1–23.
6. Martin, J. (2012). Everything you always wanted to know about the cosmological constant problem (but were afraid to ask). *Comptes Rendus Physique*, 13, 566–665. arXiv:1205.3365.
7. Padmanabhan, T. (2003). Cosmological constant — the weight of the vacuum. *Physics Reports*, 380, 235–320. arXiv:hep-th/0212290.
8. Starobinsky, A. A. (1980). A new type of isotropic cosmological models without singularity. *Physics Letters B*, 91, 99–102.
9. Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6. arXiv:1807.06209.
10. Gell-Mann, M., Oakes, R. J., & Renner, B. (1968). Behavior of current divergences under $SU(3) \times SU(3)$. *Physical Review*, 175, 2195–2199.
11. Hoferichter, M., Ruiz de Elvira, J., Hammer, B., & Meißner, U.-G. (2015). Pion-nucleon scattering: precise predictions for the pion-nucleon sigma term. *Physical Review Letters*, 115, 092301. arXiv:1506.04142.
12. Gupta, R., Jang, Y.-C., Lin, H.-W., Yoon, B., & Bhattacharya, T. (2018). Pion-nucleon sigma term from lattice QCD. *Physical Review Letters*, 127, 242002. arXiv:1812.10377.
13. Drukarev, E. G., & Levin, E. M. (1991). Structure of nuclear matter and the nucleon sigma term. *Progress in Particle and Nuclear Physics*, 27, 77–143.
14. Cohen, T. D., Furnstahl, R. J., & Griegel, D. K. (1992). Quark and gluon condensates in nuclear matter. *Physical Review C*, 45, 1881–1893.
15. Collins, J. C., Duncan, A., & Joglekar, S. D. (1977). Trace and dilatation anomalies in gauge theories. *Physical Review D*, 16, 438–449.
16. Schwinger, J. (1951). On gauge invariance and vacuum polarization. *Physical Review*, 82, 664–679.
17. DeWitt, B. S. (1965). *Dynamical Theory of Groups and Fields*. Gordon and Breach, New York.
18. Manohar, A. V., & Georgi, H. (1984). Chiral quarks and the non-relativistic quark model. *Nuclear Physics B*, 234, 189–212.

19. Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman, San Francisco.
20. Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3, 133–181.
21. Jacobson, T. (1995). Thermodynamics of spacetime: the Einstein equation of state. *Physical Review Letters*, 75, 1260–1263. arXiv:gr-qc/9504004.
22. Verlinde, E. (2011). On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011, 29. arXiv:1001.0785.
23. Boltzmann, L. (1872). Weitere Studien über das Wärmegleichgewicht unter Gasmolekülen. *Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften in Wien*, 66, 275–370.
24. Penrose, R. (1979). Singularities and time-asymmetry. In *General Relativity: An Einstein Centenary Survey*, eds. S. W. Hawking & W. Israel, Cambridge University Press, 581–638.
25. Coleman, S. (1988). Why there is nothing rather than something: a theory of the cosmological constant. *Nuclear Physics B*, 310, 643–668.
26. Kaloper, N., & Padilla, A. (2014). Sequestering the standard model vacuum energy. *Physical Review Letters*, 112, 091304. arXiv:1309.6562.
27. Volovik, G. E. (2003). *The Universe in a Helium Droplet*. Oxford University Press.
28. Solà Peracaula, J. (2013). Cosmological constant and vacuum energy: old and new ideas. *Journal of Physics: Conference Series*, 453, 012015. arXiv:1306.4540.
29. Abbott, B. P. et al. (LIGO/Virgo Collaboration) (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116, 061102. arXiv:1602.03837.
30. Abbott, B. P. et al. (LIGO/Virgo Collaboration) (2017). GW170817: observation of gravitational waves from a binary neutron star inspiral. *Physical Review Letters*, 119, 161101. arXiv:1710.05832.
31. Binnétruy, P., Bohé, A., Caprini, C., & Dufaux, J.-F. (2012). Tidal Love numbers of black holes and neutron stars. *Physical Review D*, 86, 044023.
32. van Dokkum, P. et al. (2018). A galaxy lacking dark matter. *Nature*, 555, 629–632.
33. van Dokkum, P. et al. (2019). A second galaxy missing dark matter in the NGC 1052 group. *The Astrophysical Journal Letters*, 874, L5.
34. Di Valentino, E., Melchiorri, A., & Silk, J. (2020). Planck evidence for a closed universe and a possible crisis for cosmology. *Nature Astronomy*, 4, 196–203. arXiv:1911.02087.
35. Carrera, M., & Giulini, D. (2010). Influence of global cosmological expansion on local dynamics and kinematics. *Reviews of Modern Physics*, 82, 169–208. arXiv:0810.2712.
36. Sakharov, A. D. (1968). Vacuum quantum fluctuations in curved space and the theory of gravitation. *Soviet Physics — Doklady*, 12, 1040–1041.
37. Noether, E. (1918). Invariante Variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen*, 235–257.
38. Perrin, J. (1909). Mouvement brownien et réalité moléculaire. *Annales de Chimie et de Physique*, 18, 5–114.
39. LiteBIRD Collaboration (2023). Probing cosmic inflation with the LiteBIRD satellite. *Progress of Theoretical and Experimental Physics*, 2023, 042F01. arXiv:2202.02773.
40. CMB-S4 Collaboration (2022). CMB-S4 science case, reference design, and project plan. arXiv:2203.08024.

41. DESI Collaboration (2024). DESI 2024 VI: cosmological constraints from baryon acoustic oscillations. arXiv:2404.03002.
42. Kriger, B. (2026). *The Local Gravitation of Quantum Vacuum: A Unified Solution to the Dark Sector*. Information Physics Institute. doi:10.5281/zenodo.18987982.
43. Arnowitt, R., Deser, S., & Misner, C. W. (1962). The dynamics of general relativity. In *Gravitation: An Introduction to Current Research*, ed. L. Witten, Wiley, 227–265. arXiv:gr-qc/0405109.
44. Tryon, E. P. (1973). Is the universe a vacuum fluctuation? *Nature*, 246, 396–397.
45. Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D*, 23, 287–298.
46. Planck Collaboration (2020). Planck 2018 results. VIII. Gravitational lensing. *Astronomy & Astrophysics*, 641, A8. arXiv:1807.06210.
47. Planck Collaboration (2020). Planck 2018 results. VII. Isotropy and statistics of the CMB. *Astronomy & Astrophysics*, 641, A7. arXiv:1906.02552.
48. Poincaré, H. (1890). Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13, 1–270.
49. Kashlinsky, A., Atrio-Barandela, F., Kocevski, D., & Ebeling, H. (2008). A measurement of large-scale peculiar velocities of clusters of galaxies: results and cosmological implications. *The Astrophysical Journal Letters*, 686, L49–L52. arXiv:0809.3733.
50. Geroch, R. (1967). Topology in general relativity. *Journal of Mathematical Physics*, 8, 782–786.
51. Guth, A. H. (1981). Inflationary universe: a possible solution to the horizon and flatness problems. *Physical Review D*, 23, 347–356.
52. Riess, A. G. et al. (2022). A comprehensive measurement of the local value of the Hubble constant with 1 km/s/Mpc uncertainty from the Hubble Space Telescope and the SH0ES team. *The Astrophysical Journal Letters*, 934, L7. arXiv:2112.04510.
53. Fields, B. D. (2011). The primordial lithium problem. *Annual Review of Nuclear and Particle Science*, 61, 47–68. arXiv:1203.3551.
54. Boylan-Kolchin, M., Bullock, J. S., & Kaplinghat, M. (2011). Too big to fail? The puzzling darkness of massive Milky Way subhaloes. *Monthly Notices of the Royal Astronomical Society*, 415, L40–L44. arXiv:1103.0007.
55. Moore, B. et al. (1999). Dark matter substructure within galactic halos. *The Astrophysical Journal*, 524, L19–L22. arXiv:astro-ph/9907411.
56. de Blok, W. J. G. (2010). The core-cusp problem. *Advances in Astronomy*, 2010, 789293. arXiv:0910.3538.
57. Clowe, D. et al. (2006). A direct empirical proof of the existence of dark matter. *The Astrophysical Journal Letters*, 648, L109–L113. arXiv:astro-ph/0608407.
58. Markevitch, M. et al. (2004). Direct constraints on the dark matter self-interaction cross section from the merging galaxy cluster 1E 0657-56. *The Astrophysical Journal*, 606, 819–824. arXiv:astro-ph/0309303.